



Saline Bottle Alert

PROBLEM STATEMENT

In hospitals, IV glucose bottles are manually monitored by nurses. If the bottle becomes empty and is not replaced on time, it may cause serious health risks such as air entering the bloodstream. Due to heavy workload and multiple patients, continuous manual monitoring is difficult. Therefore, there is a need for a real-time automatic monitoring system to ensure patient safety.

PROJECT OVERVIEW

Saline bottle Alert is an IoT-based healthcare monitoring system. It uses an ultrasonic sensor to measure the liquid level inside the glucose bottle. The NodeMCU processes the data and sends it to the Mobile via WiFi. The level is displayed in percentage format on a mobile application, and alerts are generated when the bottle is nearly empty.

SOLUTION OFFERED

The solution consists of a sensing module and a monitoring module. The ultrasonic sensor measures the distance between the sensor and the glucose liquid surface. The NodeMCU calculates the percentage level and uploads it to the cloud. When the level drops below 20%, a notification is sent to nurses through the mobile application. This ensures timely replacement of the IV bottle.

END USERS

- Hospitals • Nurses • Clinics • Home Patient Care

TECHNOLOGY USED

IoT Hardware: NodeMCU (ESP8266), HC-SR04 Ultrasonic Sensor, Jumper Wires. Software: Arduino IDE (Embedded C Programming), Communication: WiFi-based real-time data transmission